Malaria and Climate GAM Analysis

Automated Report

2026-03-31

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1 Overview

This report reproduces the malaria-climate generalized additive model (GAM) workflow:

- Loads and prepares monthly malaria and climate data.
- Creates lagged climate predictors.
- Fits a GAM with smooth terms.
- Reports diagnostics and concurvity.
- Exports smooth-term plots to the `Smooth Term Plots` folder.

2 Packages and Setup

```r
install_if_missing <- function(pkgs) {
  to_install <- pkgs[!pkgs %in% rownames(installed.packages())]
  if (length(to_install)) install.packages(to_install, dependencies = TRUE)
}

pkgs <- c("mgcv", "dplyr", "tidyr", "gratia", "ggplot2", "stringr", "purrr")
install_if_missing(pkgs)
invisible(lapply(pkgs, library, character.only = TRUE))

set.seed(42)
plots_dir <- file.path(getwd(), "Smooth Term Plots")
if (!dir.exists(plots_dir)) dir.create(plots_dir)
```

3 Helper Functions

```r
clean_name <- function(x) {
  x |>
    stringr::str_replace_all("s\\(|\\)", "") |>
    stringr::str_replace_all("[^A-Za-z0-9_]+", "_") |>
    stringr::str_replace("^_+|_+$", "")
}

pub_theme <- ggplot2::theme_bw(base_size = 12) +
  ggplot2::theme(
    panel.grid = ggplot2::element_blank(),
    plot.title = ggplot2::element_text(face = "bold", hjust = 0),
    axis.title = ggplot2::element_text(face = "bold"),
    axis.text = ggplot2::element_text(color = "black"),
    strip.text = ggplot2::element_text(face = "bold"),
    legend.position = "none"
  )

safe_concurvity <- function(fit) {
  out <- try(mgcv::concurvity(fit, full = TRUE), silent = TRUE)
  if (inherits(out, "try-error")) mgcv::concurvity(fit, full = FALSE) else out
}

pretty_labs <- c(
  daytime_temperature = "Daytime Temperature (deg C)",
  nighttime_temperature = "Nighttime Temperature (deg C)",
  monthly_rainfall = "Monthly Rainfall (mm)",
  Relative_humidity = "Relative Humidity (%)",
  lag1_daytime_temperature = "Day Temp (lag 1 mo, deg C)",
  lag2_daytime_temperature = "Day Temp (lag 2 mo, deg C)",
  lag3_daytime_temperature = "Day Temp (lag 3 mo, deg C)",
  lag1_nighttime_temperature = "Night Temp (lag 1 mo, deg C)",
  lag2_nighttime_temperature = "Night Temp (lag 2 mo, deg C)",
  lag3_nighttime_temperature = "Night Temp (lag 3 mo, deg C)",
  lag1_monthly_rainfall = "Rainfall (lag 1 mo, mm)",
  lag2_monthly_rainfall = "Rainfall (lag 2 mo, mm)",
```

```r
  lag3_monthly_rainfall = "Rainfall (lag 3 mo, mm)",
  lag1_Relative_humidity = "Rel. Humidity (lag 1 mo, %)",
  lag2_Relative_humidity = "Rel. Humidity (lag 2 mo, %)",
  lag3_Relative_humidity = "Rel. Humidity (lag 3 mo, %)",
  yearmon = "Time"
)

draw_one_smooth <- function(fit, s_label) {
  var_raw <- clean_name(s_label)
  x_lab <- pretty_labs[[var_raw]]
  if (is.null(x_lab)) x_lab <- var_raw

  gratia::draw(fit, select = s_label, residuals = FALSE, ci_col = "grey50") +
    ggplot2::labs(
      title = paste0("Smooth of ", x_lab),
      x = x_lab,
      y = "Partial effect on Malaria incidence per 10,000"
    ) +
    pub_theme
}
```

4 Load and Prepare Data

```r
dat <- read.csv("Data.csv", stringsAsFactors = FALSE)
dat$yearmon <- as.numeric(as.Date(paste0(dat$yearmon, "-01")))

dat <- dat %>%
  dplyr::arrange(yearmon) %>%
  dplyr::mutate(
    lag1_daytime_temperature = dplyr::lag(daytime_temperature, 1),
    lag2_daytime_temperature = dplyr::lag(daytime_temperature, 2),
    lag3_daytime_temperature = dplyr::lag(daytime_temperature, 3),
    lag1_nighttime_temperature = dplyr::lag(nighttime_temperature, 1),
    lag2_nighttime_temperature = dplyr::lag(nighttime_temperature, 2),
    lag3_nighttime_temperature = dplyr::lag(nighttime_temperature, 3),
    lag1_monthly_rainfall = dplyr::lag(monthly_rainfall, 1),
    lag2_monthly_rainfall = dplyr::lag(monthly_rainfall, 2),
    lag3_monthly_rainfall = dplyr::lag(monthly_rainfall, 3),
    lag1_Relative_humidity = dplyr::lag(Relative_humidity, 1),
    lag2_Relative_humidity = dplyr::lag(Relative_humidity, 2),
    lag3_Relative_humidity = dplyr::lag(Relative_humidity, 3)
  ) %>%
  tidyr::drop_na()

dplyr::glimpse(dat)
```

```
## Rows: 127
## Columns: 30
## $ Month                       <chr> "Apr", "May", "Jun", "Jul", "Aug", "Sep", ~
## $ Year                        <int> 2014, 2014, 2014, 2014, 2014, 2014, 2014, ~
```

```
## $ yearmon                     <dbl> 16161, 16191, 16222, 16252, 16283, 16314, ~
## $ Location                    <chr> "Dar-es-salaam", "Dar-es-salaam", "Dar-es-~
## $ X.5_years_OPD               <int> 42675, 51606, 44113, 36166, 38248, 36672, ~
## $ X.5_years_OPD.1             <int> 26295, 29828, 27512, 24984, 23506, 21971, ~
## $ Total_cases_OPD             <int> 68970, 81434, 71625, 61150, 61754, 58643, ~
## $ RCH                         <int> 490, 568, 582, 820, 714, 441, 457, 1199, 5~
## $ X.5_years_IPD               <int> 1318, 1401, 1240, 923, 707, 1014, 1059, 12~
## $ All_cases                   <int> 70778, 83403, 73447, 62893, 63175, 60098, ~
## $ Population                  <int> 4622518, 4622518, 4622518, 4622518, 462251~
## $ Malaria_incidence_per_10000 <dbl> 153.11568, 180.42764, 158.88959, 136.05788~
## $ daytime_temperature         <dbl> 31.4, 30.6, 31.1, 30.4, 31.8, 30.6, 32.6, ~
## $ nighttime_temperature       <dbl> 23.5, 22.4, 20.9, 19.8, 20.0, 20.2, 21.7, ~
## $ Relative_humidity           <int> 90, 88, 84, 82, 80, 80, 78, 78, 77, 76, 74~
## $ monthly_rainfall            <dbl> 449.2, 154.5, 4.2, 7.9, 0.5, 60.8, 20.1, 3~
## $ Anomalies_rainfall          <dbl> 1.4027035, -0.1960831, -0.5062633, -0.5373~
## $ Anomalies_relative_humidity <dbl> 0.97435470, 0.87432881, 1.11174696, 1.3435~
## $ lag1_daytime_temperature    <dbl> 32.3, 31.4, 30.6, 31.1, 30.4, 31.8, 30.6, ~
## $ lag2_daytime_temperature    <dbl> 32.5, 32.3, 31.4, 30.6, 31.1, 30.4, 31.8, ~
## $ lag3_daytime_temperature    <dbl> 33.4, 32.5, 32.3, 31.4, 30.6, 31.1, 30.4, ~
## $ lag1_nighttime_temperature  <dbl> 23.7, 23.5, 22.4, 20.9, 19.8, 20.0, 20.2, ~
## $ lag2_nighttime_temperature  <dbl> 24.4, 23.7, 23.5, 22.4, 20.9, 19.8, 20.0, ~
## $ lag3_nighttime_temperature  <dbl> 25.9, 24.4, 23.7, 23.5, 22.4, 20.9, 19.8, ~
## $ lag1_monthly_rainfall       <dbl> 426.3, 449.2, 154.5, 4.2, 7.9, 0.5, 60.8, ~
## $ lag2_monthly_rainfall       <dbl> 80.6, 426.3, 449.2, 154.5, 4.2, 7.9, 0.5, ~
## $ lag3_monthly_rainfall       <dbl> 0.9, 80.6, 426.3, 449.2, 154.5, 4.2, 7.9, ~
## $ lag1_Relative_humidity      <int> 87, 90, 88, 84, 82, 80, 80, 78, 78, 77, 76~
## $ lag2_Relative_humidity      <int> 81, 87, 90, 88, 84, 82, 80, 80, 78, 78, 77~
## $ lag3_Relative_humidity      <int> 76, 81, 87, 90, 88, 84, 82, 80, 80, 78, 78~
```

5 Fit GAM

```r
gam_fit <- mgcv::gam(
  Malaria_incidence_per_10000 ~
    s(daytime_temperature, bs = "cs", k = 10) +
    s(nighttime_temperature, bs = "cs", k = 10) +
    s(monthly_rainfall, bs = "cs", k = 10) +
    s(Relative_humidity, bs = "cs", k = 10) +
    s(lag1_daytime_temperature, bs = "cs", k = 10) +
    s(lag2_daytime_temperature, bs = "cs", k = 10) +
    s(lag3_daytime_temperature, bs = "cs", k = 10) +
    s(lag1_nighttime_temperature, bs = "cs", k = 10) +
    s(lag2_nighttime_temperature, bs = "cs", k = 10) +
    s(lag3_nighttime_temperature, bs = "cs", k = 10) +
    s(lag1_monthly_rainfall, bs = "cs", k = 10) +
    s(lag2_monthly_rainfall, bs = "cs", k = 10) +
    s(lag3_monthly_rainfall, bs = "cs", k = 10) +
    s(lag1_Relative_humidity, bs = "cs", k = 10) +
    s(lag2_Relative_humidity, bs = "cs", k = 10) +
    s(lag3_Relative_humidity, bs = "cs", k = 10) +
    s(yearmon, bs = "cs", k = 12),
  data = dat,
```

```r
  method = "REML",
  select = TRUE
)
```

6 Model Diagnostics

6.1 GAM Summary

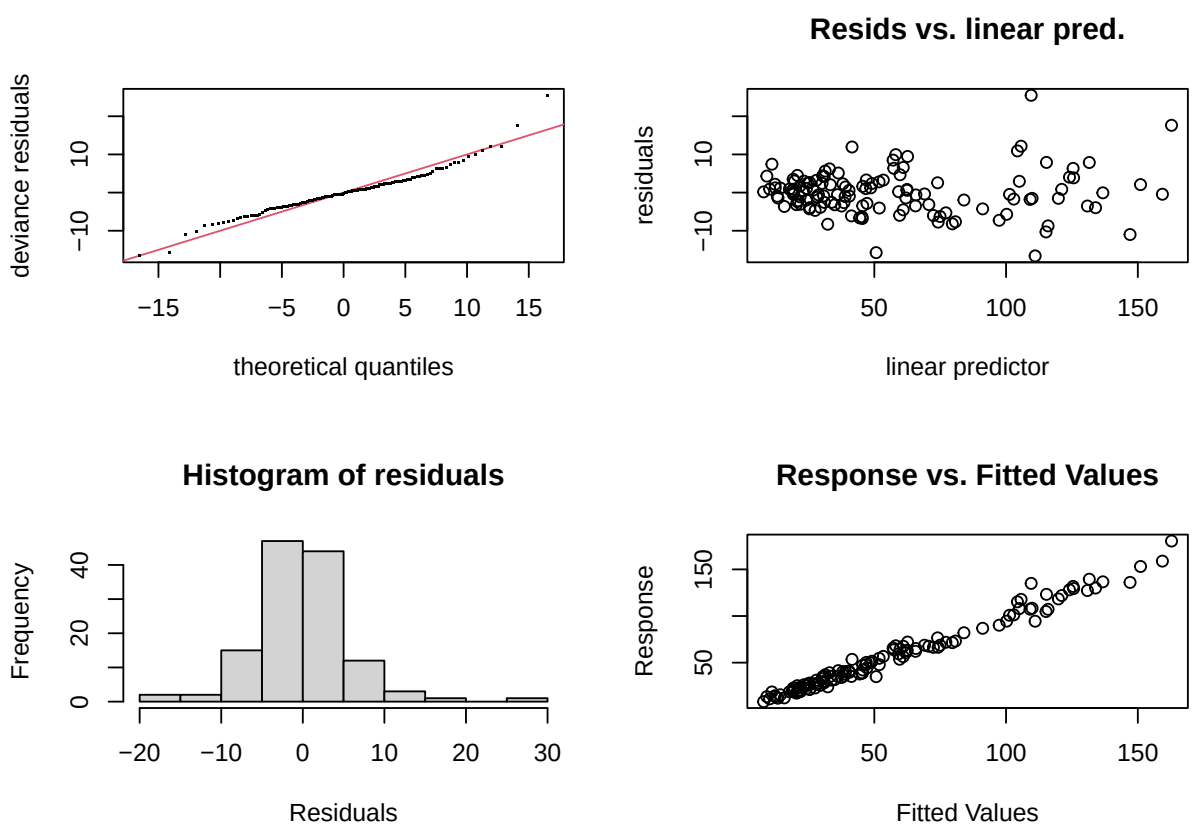
```r
summary(gam_fit)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## Malaria_incidence_per_10000 ~ s(daytime_temperature, bs = "cs",
##     k = 10) + s(nighttime_temperature, bs = "cs", k = 10) + s(monthly_rainfall,
##     bs = "cs", k = 10) + s(Relative_humidity, bs = "cs", k = 10) +
##     s(lag1_daytime_temperature, bs = "cs", k = 10) + s(lag2_daytime_temperature,
##     bs = "cs", k = 10) + s(lag3_daytime_temperature, bs = "cs",
##     k = 10) + s(lag1_nighttime_temperature, bs = "cs", k = 10) +
##     s(lag2_nighttime_temperature, bs = "cs", k = 10) + s(lag3_nighttime_temperature,
##     bs = "cs", k = 10) + s(lag1_monthly_rainfall, bs = "cs",
##     k = 10) + s(lag2_monthly_rainfall, bs = "cs", k = 10) + s(lag3_monthly_rainfall,
##     bs = "cs", k = 10) + s(lag1_Relative_humidity, bs = "cs",
##     k = 10) + s(lag2_Relative_humidity, bs = "cs", k = 10) +
##     s(lag3_Relative_humidity, bs = "cs", k = 10) + s(yearmon,
##     bs = "cs", k = 12)
##
## Parametric coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  56.8783     0.5517   103.1   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                                    edf Ref.df       F  p-value
## s(daytime_temperature)        1.454e+00      9   0.886  0.00693 **
## s(nighttime_temperature)      1.473e-05      9   0.000  0.62809
## s(monthly_rainfall)           3.526e-05      9   0.000  0.60748
## s(Relative_humidity)          2.093e+00      9   0.928  0.01165 *
## s(lag1_daytime_temperature)   3.428e+00      9   2.501  0.00020 ***
## s(lag2_daytime_temperature)   4.016e-05      9   0.000  0.96159
## s(lag3_daytime_temperature)   4.598e-04      9   0.000  0.36667
## s(lag1_nighttime_temperature) 4.227e-05      9   0.000  0.74812
## s(lag2_nighttime_temperature) 1.288e+00      9   0.385  0.02412 *
## s(lag3_nighttime_temperature) 3.955e-01      9   0.064  0.14437
## s(lag1_monthly_rainfall)      1.810e+00      9   2.182 2.85e-05 ***
## s(lag2_monthly_rainfall)      1.316e+00      9   2.517 1.23e-05 ***
## s(lag3_monthly_rainfall)      6.373e-01      9   0.170  0.10235
```

```
## s(lag1_Relative_humidity)     3.499e-05      9   0.000  0.85551
## s(lag2_Relative_humidity)     4.727e-05      9   0.000  0.96977
## s(lag3_Relative_humidity)     2.795e-05      9   0.000  0.70983
## s(yearmon)                    7.633e+00     11 418.094  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.975   Deviance explained = 97.9%
## -REML = 442.86  Scale est. = 38.651    n = 127
```

6.2 Basis Dimension Check

```r
gam.check(gam_fit)
```




```
##
## Method: REML   Optimizer: outer newton
## full convergence after 12 iterations.
## Gradient range [-7.406122e-05,0.0001730586]
## (score 442.8629 & scale 38.6515).
## Hessian positive definite, eigenvalue range [5.190059e-06,63.33833].
## Model rank =  156 / 156
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
```

```
## indicate that k is too low, especially if edf is close to k'.
##
##                                      k'      edf k-index p-value
## s(daytime_temperature)        9.00e+00 1.45e+00    1.09   0.805
## s(nighttime_temperature)      9.00e+00 1.47e-05    0.95   0.285
## s(monthly_rainfall)           9.00e+00 3.53e-05    1.11   0.855
## s(Relative_humidity)          9.00e+00 2.09e+00    1.16   0.945
## s(lag1_daytime_temperature)   9.00e+00 3.43e+00    1.01   0.610
## s(lag2_daytime_temperature)   9.00e+00 4.02e-05    1.11   0.855
## s(lag3_daytime_temperature)   9.00e+00 4.60e-04    1.02   0.610
## s(lag1_nighttime_temperature) 9.00e+00 4.23e-05    0.97   0.255
## s(lag2_nighttime_temperature) 9.00e+00 1.29e+00    0.99   0.500
## s(lag3_nighttime_temperature) 9.00e+00 3.95e-01    1.15   0.945
## s(lag1_monthly_rainfall)      9.00e+00 1.81e+00    0.94   0.245
## s(lag2_monthly_rainfall)      9.00e+00 1.32e+00    1.14   0.935
## s(lag3_monthly_rainfall)      9.00e+00 6.37e-01    1.04   0.660
## s(lag1_Relative_humidity)     9.00e+00 3.50e-05    1.06   0.665
## s(lag2_Relative_humidity)     9.00e+00 4.73e-05    0.96   0.275
## s(lag3_Relative_humidity)     9.00e+00 2.80e-05    0.84   0.025 *
## s(yearmon)                    1.10e+01 7.63e+00    0.89   0.110
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

6.3 Concurvity

```r
safe_concurvity(gam_fit)
```

```
## $worst
##                                     para s(daytime_temperature)
## para                          1.000000e+00           6.166734e-31
## s(daytime_temperature)        5.714583e-31           1.000000e+00
## s(nighttime_temperature)      1.105090e-31           6.464963e-01
## s(monthly_rainfall)           4.250997e-32           3.185354e-01
## s(Relative_humidity)          8.868680e-31           2.941828e-01
## s(lag1_daytime_temperature)   2.903150e-32           5.246637e-01
## s(lag2_daytime_temperature)   4.849040e-31           5.979925e-01
## s(lag3_daytime_temperature)   4.985126e-32           5.727969e-01
## s(lag1_nighttime_temperature) 4.566425e-32           4.075802e-01
## s(lag2_nighttime_temperature) 1.736021e-31           2.456582e-01
## s(lag3_nighttime_temperature) 1.015921e-31           2.517603e-01
## s(lag1_monthly_rainfall)      1.821293e-31           1.651832e-01
## s(lag2_monthly_rainfall)      1.548751e-31           2.383678e-01
## s(lag3_monthly_rainfall)      3.728494e-31           2.621794e-01
## s(lag1_Relative_humidity)     2.136175e-31           3.108126e-01
## s(lag2_Relative_humidity)     2.669491e-31           3.896440e-01
## s(lag3_Relative_humidity)     1.034167e-30           3.890495e-01
## s(yearmon)                    3.201108e-31           4.545514e-01
##                               s(nighttime_temperature) s(monthly_rainfall)
## para                                      1.947291e-31        4.297797e-32
## s(daytime_temperature)                    6.464963e-01        3.185354e-01
## s(nighttime_temperature)                  1.000000e+00        6.382538e-01
## s(monthly_rainfall)                       6.382538e-01        1.000000e+00
```

```
## s(Relative_humidity)                      4.327093e-01        4.380613e-01
## s(lag1_daytime_temperature)               6.886362e-01        5.150445e-01
## s(lag2_daytime_temperature)               4.972805e-01        3.931732e-01
## s(lag3_daytime_temperature)               2.832003e-01        3.072931e-01
## s(lag1_nighttime_temperature)             7.396892e-01        4.863500e-01
## s(lag2_nighttime_temperature)             3.489941e-01        3.040753e-01
## s(lag3_nighttime_temperature)             4.223008e-01        2.627590e-01
## s(lag1_monthly_rainfall)                  3.600858e-01        3.099688e-01
## s(lag2_monthly_rainfall)                  2.623637e-01        2.253588e-01
## s(lag3_monthly_rainfall)                  2.469487e-01        2.079886e-01
## s(lag1_Relative_humidity)                 3.567588e-01        3.426382e-01
## s(lag2_Relative_humidity)                 2.542011e-01        2.951983e-01
## s(lag3_Relative_humidity)                 4.394143e-01        2.826056e-01
## s(yearmon)                                2.634103e-01        2.226559e-01
##                                s(Relative_humidity) s(lag1_daytime_temperature)
## para                                   7.465747e-31                2.385658e-32
## s(daytime_temperature)                 2.941828e-01                5.246637e-01
## s(nighttime_temperature)               4.327093e-01                6.886362e-01
## s(monthly_rainfall)                    4.380613e-01                5.150445e-01
## s(Relative_humidity)                   1.000000e+00                2.366341e-01
## s(lag1_daytime_temperature)            2.366341e-01                1.000000e+00
## s(lag2_daytime_temperature)            3.576058e-01                5.275223e-01
## s(lag3_daytime_temperature)            3.583173e-01                5.969929e-01
## s(lag1_nighttime_temperature)          3.220922e-01                6.367968e-01
## s(lag2_nighttime_temperature)          3.133775e-01                4.069424e-01
## s(lag3_nighttime_temperature)          4.182463e-01                2.348572e-01
## s(lag1_monthly_rainfall)               2.698726e-01                3.126666e-01
## s(lag2_monthly_rainfall)               2.078099e-01                1.739700e-01
## s(lag3_monthly_rainfall)               2.052024e-01                2.356993e-01
## s(lag1_Relative_humidity)              4.456855e-01                2.962968e-01
## s(lag2_Relative_humidity)              2.726217e-01                3.107844e-01
## s(lag3_Relative_humidity)              2.957629e-01                3.889979e-01
## s(yearmon)                             4.153012e-01                4.656438e-01
##                                s(lag2_daytime_temperature)
## para                                        2.116167e-31
## s(daytime_temperature)                      5.979925e-01
## s(nighttime_temperature)                    4.972805e-01
## s(monthly_rainfall)                         3.931732e-01
## s(Relative_humidity)                        3.576058e-01
## s(lag1_daytime_temperature)                 5.275223e-01
## s(lag2_daytime_temperature)                 1.000000e+00
## s(lag3_daytime_temperature)                 5.278104e-01
## s(lag1_nighttime_temperature)               6.878674e-01
## s(lag2_nighttime_temperature)               6.349677e-01
## s(lag3_nighttime_temperature)               4.108411e-01
## s(lag1_monthly_rainfall)                    5.333284e-01
## s(lag2_monthly_rainfall)                    3.257549e-01
## s(lag3_monthly_rainfall)                    1.651894e-01
## s(lag1_Relative_humidity)                   2.457624e-01
## s(lag2_Relative_humidity)                   3.026766e-01
## s(lag3_Relative_humidity)                   3.093434e-01
## s(yearmon)                                  4.878732e-01
##                                s(lag3_daytime_temperature)
## para                                        9.468537e-32
```

```
## s(daytime_temperature)                      5.727969e-01
## s(nighttime_temperature)                    2.832003e-01
## s(monthly_rainfall)                         3.072931e-01
## s(Relative_humidity)                        3.583173e-01
## s(lag1_daytime_temperature)                 5.969929e-01
## s(lag2_daytime_temperature)                 5.278104e-01
## s(lag3_daytime_temperature)                 1.000000e+00
## s(lag1_nighttime_temperature)               4.940603e-01
## s(lag2_nighttime_temperature)               6.870758e-01
## s(lag3_nighttime_temperature)               6.305484e-01
## s(lag1_monthly_rainfall)                    4.219852e-01
## s(lag2_monthly_rainfall)                    5.279393e-01
## s(lag3_monthly_rainfall)                    3.215775e-01
## s(lag1_Relative_humidity)                   3.695971e-01
## s(lag2_Relative_humidity)                   2.403401e-01
## s(lag3_Relative_humidity)                   2.925929e-01
## s(yearmon)                                  5.066716e-01
##                              s(lag1_nighttime_temperature)
## para                                          5.400488e-32
## s(daytime_temperature)                        4.075802e-01
## s(nighttime_temperature)                      7.396892e-01
## s(monthly_rainfall)                           4.863500e-01
## s(Relative_humidity)                          3.220922e-01
## s(lag1_daytime_temperature)                   6.367968e-01
## s(lag2_daytime_temperature)                   6.878674e-01
## s(lag3_daytime_temperature)                   4.940603e-01
## s(lag1_nighttime_temperature)                 1.000000e+00
## s(lag2_nighttime_temperature)                 7.374020e-01
## s(lag3_nighttime_temperature)                 3.495933e-01
## s(lag1_monthly_rainfall)                      6.634802e-01
## s(lag2_monthly_rainfall)                      3.570172e-01
## s(lag3_monthly_rainfall)                      2.746729e-01
## s(lag1_Relative_humidity)                     4.416982e-01
## s(lag2_Relative_humidity)                     3.505553e-01
## s(lag3_Relative_humidity)                     2.635228e-01
## s(yearmon)                                    2.962516e-01
##                              s(lag2_nighttime_temperature)
## para                                          9.017240e-32
## s(daytime_temperature)                        2.456582e-01
## s(nighttime_temperature)                      3.489941e-01
## s(monthly_rainfall)                           3.040753e-01
## s(Relative_humidity)                          3.133775e-01
## s(lag1_daytime_temperature)                   4.069424e-01
## s(lag2_daytime_temperature)                   6.349677e-01
## s(lag3_daytime_temperature)                   6.870758e-01
## s(lag1_nighttime_temperature)                 7.374020e-01
## s(lag2_nighttime_temperature)                 1.000000e+00
## s(lag3_nighttime_temperature)                 7.357457e-01
## s(lag1_monthly_rainfall)                      5.114232e-01
## s(lag2_monthly_rainfall)                      6.622938e-01
## s(lag3_monthly_rainfall)                      3.489401e-01
## s(lag1_Relative_humidity)                     3.215640e-01
## s(lag2_Relative_humidity)                     4.288186e-01
## s(lag3_Relative_humidity)                     3.467456e-01
```

```
## s(yearmon)                                    2.762036e-01
##                                s(lag3_nighttime_temperature)
## para                                          5.505519e-32
## s(daytime_temperature)                        2.517603e-01
## s(nighttime_temperature)                      4.223008e-01
## s(monthly_rainfall)                           2.627590e-01
## s(Relative_humidity)                          4.182463e-01
## s(lag1_daytime_temperature)                   2.348572e-01
## s(lag2_daytime_temperature)                   4.108411e-01
## s(lag3_daytime_temperature)                   6.305484e-01
## s(lag1_nighttime_temperature)                 3.495933e-01
## s(lag2_nighttime_temperature)                 7.357457e-01
## s(lag3_nighttime_temperature)                 1.000000e+00
## s(lag1_monthly_rainfall)                      3.288810e-01
## s(lag2_monthly_rainfall)                      5.098897e-01
## s(lag3_monthly_rainfall)                      6.415439e-01
## s(lag1_Relative_humidity)                     3.230439e-01
## s(lag2_Relative_humidity)                     3.153820e-01
## s(lag3_Relative_humidity)                     4.227684e-01
## s(yearmon)                                    2.657253e-01
##                                s(lag1_monthly_rainfall) s(lag2_monthly_rainfall)
## para                                         3.747264e-32             1.559317e-31
## s(daytime_temperature)                       1.651832e-01             2.383678e-01
## s(nighttime_temperature)                     3.600858e-01             2.623637e-01
## s(monthly_rainfall)                          3.099688e-01             2.253588e-01
## s(Relative_humidity)                         2.698726e-01             2.078099e-01
## s(lag1_daytime_temperature)                  3.126666e-01             1.739700e-01
## s(lag2_daytime_temperature)                  5.333284e-01             3.257549e-01
## s(lag3_daytime_temperature)                  4.219852e-01             5.279393e-01
## s(lag1_nighttime_temperature)                6.634802e-01             3.570172e-01
## s(lag2_nighttime_temperature)                5.114232e-01             6.622938e-01
## s(lag3_nighttime_temperature)                3.288810e-01             5.098897e-01
## s(lag1_monthly_rainfall)                     1.000000e+00             3.208239e-01
## s(lag2_monthly_rainfall)                     3.208239e-01             1.000000e+00
## s(lag3_monthly_rainfall)                     2.108976e-01             3.147362e-01
## s(lag1_Relative_humidity)                    4.588327e-01             2.679763e-01
## s(lag2_Relative_humidity)                    3.219752e-01             4.548275e-01
## s(lag3_Relative_humidity)                    2.857721e-01             3.226034e-01
## s(yearmon)                                   2.263586e-01             2.562146e-01
##                                s(lag3_monthly_rainfall)
## para                                         7.178911e-32
## s(daytime_temperature)                       2.621794e-01
## s(nighttime_temperature)                     2.469487e-01
## s(monthly_rainfall)                          2.079886e-01
## s(Relative_humidity)                         2.052024e-01
## s(lag1_daytime_temperature)                  2.356993e-01
## s(lag2_daytime_temperature)                  1.651894e-01
## s(lag3_daytime_temperature)                  3.215775e-01
## s(lag1_nighttime_temperature)                2.746729e-01
## s(lag2_nighttime_temperature)                3.489401e-01
## s(lag3_nighttime_temperature)                6.415439e-01
## s(lag1_monthly_rainfall)                     2.108976e-01
## s(lag2_monthly_rainfall)                     3.147362e-01
## s(lag3_monthly_rainfall)                     1.000000e+00
```

```
## s(lag1_Relative_humidity)                1.946728e-01
## s(lag2_Relative_humidity)                2.678851e-01
## s(lag3_Relative_humidity)                4.541746e-01
## s(yearmon)                               2.369475e-01
##                              s(lag1_Relative_humidity)
## para                                      7.513207e-32
## s(daytime_temperature)                    3.108126e-01
## s(nighttime_temperature)                  3.567588e-01
## s(monthly_rainfall)                       3.426382e-01
## s(Relative_humidity)                      4.456855e-01
## s(lag1_daytime_temperature)               2.962968e-01
## s(lag2_daytime_temperature)               2.457624e-01
## s(lag3_daytime_temperature)               3.695971e-01
## s(lag1_nighttime_temperature)             4.416982e-01
## s(lag2_nighttime_temperature)             3.215640e-01
## s(lag3_nighttime_temperature)             3.230439e-01
## s(lag1_monthly_rainfall)                  4.588327e-01
## s(lag2_monthly_rainfall)                  2.679763e-01
## s(lag3_monthly_rainfall)                  1.946728e-01
## s(lag1_Relative_humidity)                 1.000000e+00
## s(lag2_Relative_humidity)                 4.460679e-01
## s(lag3_Relative_humidity)                 2.789358e-01
## s(yearmon)                                3.920809e-01
##                              s(lag2_Relative_humidity)
## para                                      2.757957e-31
## s(daytime_temperature)                    3.896440e-01
## s(nighttime_temperature)                  2.542011e-01
## s(monthly_rainfall)                       2.951983e-01
## s(Relative_humidity)                      2.726217e-01
## s(lag1_daytime_temperature)               3.107844e-01
## s(lag2_daytime_temperature)               3.026766e-01
## s(lag3_daytime_temperature)               2.403401e-01
## s(lag1_nighttime_temperature)             3.505553e-01
## s(lag2_nighttime_temperature)             4.288186e-01
## s(lag3_nighttime_temperature)             3.153820e-01
## s(lag1_monthly_rainfall)                  3.219752e-01
## s(lag2_monthly_rainfall)                  4.548275e-01
## s(lag3_monthly_rainfall)                  2.678851e-01
## s(lag1_Relative_humidity)                 4.460679e-01
## s(lag2_Relative_humidity)                 1.000000e+00
## s(lag3_Relative_humidity)                 4.423507e-01
## s(yearmon)                                3.766419e-01
##                              s(lag3_Relative_humidity)   s(yearmon)
## para                                      1.333241e-30 3.931064e-31
## s(daytime_temperature)                    3.890495e-01 4.545514e-01
## s(nighttime_temperature)                  4.394143e-01 2.634103e-01
## s(monthly_rainfall)                       2.826056e-01 2.226559e-01
## s(Relative_humidity)                      2.957629e-01 4.153012e-01
## s(lag1_daytime_temperature)               3.889979e-01 4.656438e-01
## s(lag2_daytime_temperature)               3.093434e-01 4.878732e-01
## s(lag3_daytime_temperature)               2.925929e-01 5.066716e-01
## s(lag1_nighttime_temperature)             2.635228e-01 2.962516e-01
## s(lag2_nighttime_temperature)             3.467456e-01 2.762036e-01
## s(lag3_nighttime_temperature)             4.227684e-01 2.657253e-01
```

```
## s(lag1_monthly_rainfall)                  2.857721e-01 2.263586e-01
## s(lag2_monthly_rainfall)                  3.226034e-01 2.562146e-01
## s(lag3_monthly_rainfall)                  4.541746e-01 2.369475e-01
## s(lag1_Relative_humidity)                 2.789358e-01 3.920809e-01
## s(lag2_Relative_humidity)                 4.423507e-01 3.766419e-01
## s(lag3_Relative_humidity)                 1.000000e+00 3.102689e-01
## s(yearmon)                                3.102689e-01 1.000000e+00
##
## $observed
##                                     para s(daytime_temperature)
## para                          1.000000e+00           5.410502e-32
## s(daytime_temperature)        5.714583e-31           1.000000e+00
## s(nighttime_temperature)      1.105090e-31           4.270953e-01
## s(monthly_rainfall)           4.250997e-32           1.893196e-01
## s(Relative_humidity)          8.868680e-31           2.069223e-01
## s(lag1_daytime_temperature)   2.903150e-32           4.614963e-01
## s(lag2_daytime_temperature)   4.849040e-31           3.059246e-01
## s(lag3_daytime_temperature)   4.985126e-32           2.468477e-01
## s(lag1_nighttime_temperature) 4.566425e-32           2.730532e-01
## s(lag2_nighttime_temperature) 1.736021e-31           1.079159e-01
## s(lag3_nighttime_temperature) 1.015921e-31           7.153410e-02
## s(lag1_monthly_rainfall)      1.821293e-31           1.074834e-01
## s(lag2_monthly_rainfall)      1.548751e-31           1.154708e-01
## s(lag3_monthly_rainfall)      3.728494e-31           1.163275e-01
## s(lag1_Relative_humidity)     2.136175e-31           1.712653e-01
## s(lag2_Relative_humidity)     2.669491e-31           1.870100e-01
## s(lag3_Relative_humidity)     1.034167e-30           1.719192e-01
## s(yearmon)                    3.201108e-31           1.649877e-01
##                               s(nighttime_temperature) s(monthly_rainfall)
## para                                   4.327372e-32        9.683940e-33
## s(daytime_temperature)                 5.388031e-01        8.903352e-02
## s(nighttime_temperature)               1.000000e+00        4.563650e-01
## s(monthly_rainfall)                    5.356484e-01        1.000000e+00
## s(Relative_humidity)                   2.092462e-01        4.102443e-01
## s(lag1_daytime_temperature)            6.416268e-01        2.720844e-01
## s(lag2_daytime_temperature)            4.768233e-01        2.019864e-01
## s(lag3_daytime_temperature)            1.744759e-01        1.887612e-01
## s(lag1_nighttime_temperature)          7.099429e-01        2.361328e-01
## s(lag2_nighttime_temperature)          3.025214e-01        2.099904e-01
## s(lag3_nighttime_temperature)          1.087656e-01        2.226067e-01
## s(lag1_monthly_rainfall)               3.241319e-01        2.109550e-01
## s(lag2_monthly_rainfall)               1.729635e-01        1.359880e-01
## s(lag3_monthly_rainfall)               1.624181e-01        5.322879e-02
## s(lag1_Relative_humidity)              1.594814e-01        2.996672e-01
## s(lag2_Relative_humidity)              2.187244e-01        6.125405e-02
## s(lag3_Relative_humidity)              3.271746e-01        1.363512e-01
## s(yearmon)                             9.594902e-02        1.104415e-01
##                               s(Relative_humidity) s(lag1_daytime_temperature)
## para                                 1.075608e-31                7.576241e-34
## s(daytime_temperature)               8.774493e-02                3.450068e-01
## s(nighttime_temperature)             2.525419e-01                3.513998e-01
## s(monthly_rainfall)                  3.654839e-01                2.287169e-01
## s(Relative_humidity)                 1.000000e+00                1.319108e-01
## s(lag1_daytime_temperature)          1.346313e-01                1.000000e+00
```

```
## s(lag2_daytime_temperature)          2.523117e-01                3.138839e-01
## s(lag3_daytime_temperature)          1.988095e-01                1.414445e-01
## s(lag1_nighttime_temperature)        2.667283e-01                4.038466e-01
## s(lag2_nighttime_temperature)        2.141115e-01                2.900532e-01
## s(lag3_nighttime_temperature)        2.226774e-01                1.028094e-01
## s(lag1_monthly_rainfall)             1.533343e-01                8.935512e-02
## s(lag2_monthly_rainfall)             1.028455e-01                1.161664e-01
## s(lag3_monthly_rainfall)             3.318938e-02                1.274551e-01
## s(lag1_Relative_humidity)            2.854646e-01                2.468411e-01
## s(lag2_Relative_humidity)            3.317780e-02                1.351910e-01
## s(lag3_Relative_humidity)            8.405114e-02                8.979939e-02
## s(yearmon)                           8.276900e-02                2.225296e-01
##                                s(lag2_daytime_temperature)
## para                                          9.712610e-33
## s(daytime_temperature)                        1.756542e-01
## s(nighttime_temperature)                      2.341022e-01
## s(monthly_rainfall)                           1.989807e-01
## s(Relative_humidity)                          2.980648e-01
## s(lag1_daytime_temperature)                   3.490277e-01
## s(lag2_daytime_temperature)                   1.000000e+00
## s(lag3_daytime_temperature)                   3.085988e-01
## s(lag1_nighttime_temperature)                 3.730106e-01
## s(lag2_nighttime_temperature)                 3.937937e-01
## s(lag3_nighttime_temperature)                 2.690670e-01
## s(lag1_monthly_rainfall)                      2.328955e-01
## s(lag2_monthly_rainfall)                      9.288432e-02
## s(lag3_monthly_rainfall)                      1.036461e-01
## s(lag1_Relative_humidity)                     1.294994e-01
## s(lag2_Relative_humidity)                     2.173017e-01
## s(lag3_Relative_humidity)                     1.470981e-01
## s(yearmon)                                    2.105406e-01
##                                s(lag3_daytime_temperature)
## para                                          6.942738e-35
## s(daytime_temperature)                        2.355532e-01
## s(nighttime_temperature)                      5.619032e-02
## s(monthly_rainfall)                           8.206845e-02
## s(Relative_humidity)                          1.402345e-01
## s(lag1_daytime_temperature)                   3.392905e-01
## s(lag2_daytime_temperature)                   5.099430e-01
## s(lag3_daytime_temperature)                   1.000000e+00
## s(lag1_nighttime_temperature)                 2.114215e-01
## s(lag2_nighttime_temperature)                 3.999298e-01
## s(lag3_nighttime_temperature)                 3.718227e-01
## s(lag1_monthly_rainfall)                      1.734593e-01
## s(lag2_monthly_rainfall)                      3.058680e-01
## s(lag3_monthly_rainfall)                      1.735839e-01
## s(lag1_Relative_humidity)                     1.789449e-01
## s(lag2_Relative_humidity)                     1.323483e-01
## s(lag3_Relative_humidity)                     1.765485e-01
## s(yearmon)                                    2.342656e-01
##                                s(lag1_nighttime_temperature)
## para                                            5.243615e-33
## s(daytime_temperature)                          3.342575e-01
## s(nighttime_temperature)                        7.028362e-01
```

```
## s(monthly_rainfall)                          4.032183e-01
## s(Relative_humidity)                         2.281728e-01
## s(lag1_daytime_temperature)                  5.489838e-01
## s(lag2_daytime_temperature)                  6.059469e-01
## s(lag3_daytime_temperature)                  4.667384e-01
## s(lag1_nighttime_temperature)                1.000000e+00
## s(lag2_nighttime_temperature)                7.148296e-01
## s(lag3_nighttime_temperature)                2.794982e-01
## s(lag1_monthly_rainfall)                     4.979334e-01
## s(lag2_monthly_rainfall)                     3.135509e-01
## s(lag3_monthly_rainfall)                     1.770103e-01
## s(lag1_Relative_humidity)                    1.756144e-01
## s(lag2_Relative_humidity)                    1.357374e-01
## s(lag3_Relative_humidity)                    2.172781e-01
## s(yearmon)                                   8.120665e-02
##                               s(lag2_nighttime_temperature)
## para                                         2.523756e-33
## s(daytime_temperature)                       1.786809e-01
## s(nighttime_temperature)                     2.846888e-01
## s(monthly_rainfall)                          2.268703e-01
## s(Relative_humidity)                         2.143295e-01
## s(lag1_daytime_temperature)                  3.679971e-01
## s(lag2_daytime_temperature)                  5.194863e-01
## s(lag3_daytime_temperature)                  5.099612e-01
## s(lag1_nighttime_temperature)                6.837492e-01
## s(lag2_nighttime_temperature)                1.000000e+00
## s(lag3_nighttime_temperature)                6.745460e-01
## s(lag1_monthly_rainfall)                     3.631826e-01
## s(lag2_monthly_rainfall)                     3.897551e-01
## s(lag3_monthly_rainfall)                     2.677833e-01
## s(lag1_Relative_humidity)                    1.901209e-01
## s(lag2_Relative_humidity)                    1.274159e-01
## s(lag3_Relative_humidity)                    1.065440e-01
## s(yearmon)                                   6.430551e-02
##                               s(lag3_nighttime_temperature)
## para                                         1.165837e-32
## s(daytime_temperature)                       1.647836e-01
## s(nighttime_temperature)                     6.128409e-02
## s(monthly_rainfall)                          1.366344e-01
## s(Relative_humidity)                         2.659543e-01
## s(lag1_daytime_temperature)                  1.681092e-01
## s(lag2_daytime_temperature)                  3.534375e-01
## s(lag3_daytime_temperature)                  5.427130e-01
## s(lag1_nighttime_temperature)                2.723802e-01
## s(lag2_nighttime_temperature)                7.000620e-01
## s(lag3_nighttime_temperature)                1.000000e+00
## s(lag1_monthly_rainfall)                     2.639090e-01
## s(lag2_monthly_rainfall)                     4.090156e-01
## s(lag3_monthly_rainfall)                     4.313799e-01
## s(lag1_Relative_humidity)                    2.302705e-01
## s(lag2_Relative_humidity)                    2.010401e-01
## s(lag3_Relative_humidity)                    1.505050e-01
## s(yearmon)                                   4.348438e-02
##                               s(lag1_monthly_rainfall) s(lag2_monthly_rainfall)
```

```
## para                                    1.077117e-32             6.644356e-33
## s(daytime_temperature)                  5.170862e-02             1.466065e-01
## s(nighttime_temperature)                2.416474e-01             8.993533e-02
## s(monthly_rainfall)                     1.621760e-01             1.026698e-01
## s(Relative_humidity)                    2.601397e-01             9.403021e-02
## s(lag1_daytime_temperature)             1.072768e-01             4.768310e-02
## s(lag2_daytime_temperature)             3.343320e-01             9.819926e-02
## s(lag3_daytime_temperature)             2.389176e-01             2.887425e-01
## s(lag1_nighttime_temperature)           5.176421e-01             2.267068e-01
## s(lag2_nighttime_temperature)           2.791134e-01             4.723754e-01
## s(lag3_nighttime_temperature)           2.435963e-01             2.523406e-01
## s(lag1_monthly_rainfall)                1.000000e+00             1.370878e-01
## s(lag2_monthly_rainfall)                2.402301e-01             1.000000e+00
## s(lag3_monthly_rainfall)                1.293201e-01             2.202614e-01
## s(lag1_Relative_humidity)               4.178729e-01             2.488902e-01
## s(lag2_Relative_humidity)               2.519528e-01             4.249800e-01
## s(lag3_Relative_humidity)               9.743235e-02             2.594736e-01
## s(yearmon)                              1.370209e-01             1.206949e-01
##                               s(lag3_monthly_rainfall)
## para                                    1.434123e-33
## s(daytime_temperature)                  1.431811e-01
## s(nighttime_temperature)                1.288572e-01
## s(monthly_rainfall)                     7.696523e-02
## s(Relative_humidity)                    3.884979e-02
## s(lag1_daytime_temperature)             1.408854e-01
## s(lag2_daytime_temperature)             4.846808e-02
## s(lag3_daytime_temperature)             7.641660e-02
## s(lag1_nighttime_temperature)           9.961456e-02
## s(lag2_nighttime_temperature)           2.231379e-01
## s(lag3_nighttime_temperature)           4.711924e-01
## s(lag1_monthly_rainfall)                1.031642e-01
## s(lag2_monthly_rainfall)                1.365130e-01
## s(lag3_monthly_rainfall)                1.000000e+00
## s(lag1_Relative_humidity)               9.581760e-02
## s(lag2_Relative_humidity)               2.476986e-01
## s(lag3_Relative_humidity)               4.205419e-01
## s(yearmon)                              9.521031e-02
##                               s(lag1_Relative_humidity)
## para                                     1.291777e-32
## s(daytime_temperature)                   1.498417e-01
## s(nighttime_temperature)                 1.252316e-01
## s(monthly_rainfall)                      1.668095e-01
## s(Relative_humidity)                     2.453605e-01
## s(lag1_daytime_temperature)              1.646670e-01
## s(lag2_daytime_temperature)              3.761193e-02
## s(lag3_daytime_temperature)              3.108785e-02
## s(lag1_nighttime_temperature)            1.260008e-01
## s(lag2_nighttime_temperature)            7.612981e-02
## s(lag3_nighttime_temperature)            5.853344e-02
## s(lag1_monthly_rainfall)                 1.437818e-01
## s(lag2_monthly_rainfall)                 1.361589e-01
## s(lag3_monthly_rainfall)                 8.988527e-02
## s(lag1_Relative_humidity)                1.000000e+00
## s(lag2_Relative_humidity)                2.610187e-01
```

```
## s(lag3_Relative_humidity)                  1.811335e-01
## s(yearmon)                                 3.177786e-01
##                               s(lag2_Relative_humidity)
## para                                       1.557317e-32
## s(daytime_temperature)                     3.197811e-01
## s(nighttime_temperature)                   2.099040e-01
## s(monthly_rainfall)                        1.946355e-01
## s(Relative_humidity)                       1.589287e-01
## s(lag1_daytime_temperature)                2.015994e-01
## s(lag2_daytime_temperature)                1.796222e-01
## s(lag3_daytime_temperature)                6.210625e-02
## s(lag1_nighttime_temperature)              1.956498e-01
## s(lag2_nighttime_temperature)              3.129279e-01
## s(lag3_nighttime_temperature)              2.101062e-01
## s(lag1_monthly_rainfall)                   1.162088e-01
## s(lag2_monthly_rainfall)                   3.400437e-01
## s(lag3_monthly_rainfall)                   1.701775e-01
## s(lag1_Relative_humidity)                  3.481947e-01
## s(lag2_Relative_humidity)                  1.000000e+00
## s(lag3_Relative_humidity)                  3.296838e-01
## s(yearmon)                                 2.128908e-01
##                               s(lag3_Relative_humidity)   s(yearmon)
## para                                       1.998867e-32 3.969786e-32
## s(daytime_temperature)                     1.794759e-01 1.265975e-01
## s(nighttime_temperature)                   3.339199e-01 9.839808e-02
## s(monthly_rainfall)                        2.430384e-01 9.531526e-02
## s(Relative_humidity)                       2.482392e-01 1.068238e-01
## s(lag1_daytime_temperature)                2.970338e-01 1.120332e-01
## s(lag2_daytime_temperature)                2.144917e-01 1.009889e-01
## s(lag3_daytime_temperature)                1.910527e-01 1.122732e-01
## s(lag1_nighttime_temperature)              2.033446e-01 8.485300e-02
## s(lag2_nighttime_temperature)              1.535525e-01 7.724332e-02
## s(lag3_nighttime_temperature)              2.713293e-01 7.783868e-02
## s(lag1_monthly_rainfall)                   1.922130e-01 8.790512e-02
## s(lag2_monthly_rainfall)                   1.487331e-01 6.248384e-02
## s(lag3_monthly_rainfall)                   2.778775e-01 8.737760e-02
## s(lag1_Relative_humidity)                  1.823262e-01 1.083476e-01
## s(lag2_Relative_humidity)                  3.144282e-01 1.029129e-01
## s(lag3_Relative_humidity)                  1.000000e+00 8.955518e-02
## s(yearmon)                                 1.954886e-01 1.000000e+00
##
## $estimate
##                                     para s(daytime_temperature)
## para                          1.000000e+00           1.490438e-32
## s(daytime_temperature)        5.714583e-31           1.000000e+00
## s(nighttime_temperature)      1.105090e-31           1.502813e-01
## s(monthly_rainfall)           4.250997e-32           9.488451e-02
## s(Relative_humidity)          8.868680e-31           7.093937e-02
## s(lag1_daytime_temperature)   2.903150e-32           2.026300e-01
## s(lag2_daytime_temperature)   4.849040e-31           1.259461e-01
## s(lag3_daytime_temperature)   4.985126e-32           1.285946e-01
## s(lag1_nighttime_temperature) 4.566425e-32           9.906833e-02
## s(lag2_nighttime_temperature) 1.736021e-31           1.054904e-01
## s(lag3_nighttime_temperature) 1.015921e-31           9.904589e-02
```

```
## s(lag1_monthly_rainfall)      1.821293e-31           5.224279e-02
## s(lag2_monthly_rainfall)      1.548751e-31           8.561857e-02
## s(lag3_monthly_rainfall)      3.728494e-31           9.868161e-02
## s(lag1_Relative_humidity)     2.136175e-31           1.070798e-01
## s(lag2_Relative_humidity)     2.669491e-31           1.260898e-01
## s(lag3_Relative_humidity)     1.034167e-30           1.218952e-01
## s(yearmon)                    3.201108e-31           1.158819e-01
##                               s(nighttime_temperature) s(monthly_rainfall)
## para                                     1.104075e-32        1.934243e-33
## s(daytime_temperature)                   1.614408e-01        9.072064e-02
## s(nighttime_temperature)                 1.000000e+00        1.423474e-01
## s(monthly_rainfall)                      1.645846e-01        1.000000e+00
## s(Relative_humidity)                     1.187692e-01        1.497765e-01
## s(lag1_daytime_temperature)              2.116881e-01        1.183185e-01
## s(lag2_daytime_temperature)              1.296909e-01        1.251062e-01
## s(lag3_daytime_temperature)              8.842688e-02        1.259322e-01
## s(lag1_nighttime_temperature)            2.420394e-01        1.265065e-01
## s(lag2_nighttime_temperature)            1.524208e-01        9.869528e-02
## s(lag3_nighttime_temperature)            1.240863e-01        9.395990e-02
## s(lag1_monthly_rainfall)                 1.206185e-01        9.636715e-02
## s(lag2_monthly_rainfall)                 8.133709e-02        9.492923e-02
## s(lag3_monthly_rainfall)                 8.147463e-02        7.525078e-02
## s(lag1_Relative_humidity)                1.228164e-01        1.108700e-01
## s(lag2_Relative_humidity)                1.129161e-01        9.058176e-02
## s(lag3_Relative_humidity)                1.146544e-01        8.412885e-02
## s(yearmon)                               6.032047e-02        8.692085e-02
##                               s(Relative_humidity) s(lag1_daytime_temperature)
## para                                  1.840324e-32                2.536192e-33
## s(daytime_temperature)                7.354593e-02                1.873476e-01
## s(nighttime_temperature)              1.352475e-01                2.126708e-01
## s(monthly_rainfall)                   1.327304e-01                1.360762e-01
## s(Relative_humidity)                  1.000000e+00                7.685848e-02
## s(lag1_daytime_temperature)           8.002785e-02                1.000000e+00
## s(lag2_daytime_temperature)           1.203068e-01                2.026360e-01
## s(lag3_daytime_temperature)           1.006936e-01                1.224190e-01
## s(lag1_nighttime_temperature)         1.359389e-01                1.494937e-01
## s(lag2_nighttime_temperature)         9.721067e-02                9.902901e-02
## s(lag3_nighttime_temperature)         1.053172e-01                1.035551e-01
## s(lag1_monthly_rainfall)              9.448294e-02                9.613378e-02
## s(lag2_monthly_rainfall)              8.015379e-02                5.467182e-02
## s(lag3_monthly_rainfall)              6.593974e-02                8.387076e-02
## s(lag1_Relative_humidity)             1.204277e-01                7.470085e-02
## s(lag2_Relative_humidity)             9.769597e-02                1.073568e-01
## s(lag3_Relative_humidity)             1.034553e-01                1.255257e-01
## s(yearmon)                            1.211201e-01                1.201060e-01
##                               s(lag2_daytime_temperature)
## para                                     6.542914e-33
## s(daytime_temperature)                   1.096799e-01
## s(nighttime_temperature)                 1.404734e-01
## s(monthly_rainfall)                      1.209861e-01
## s(Relative_humidity)                     1.068151e-01
## s(lag1_daytime_temperature)              1.874138e-01
## s(lag2_daytime_temperature)              1.000000e+00
## s(lag3_daytime_temperature)              2.041358e-01
```

```
## s(lag1_nighttime_temperature)                2.133615e-01
## s(lag2_nighttime_temperature)                1.562440e-01
## s(lag3_nighttime_temperature)                9.685791e-02
## s(lag1_monthly_rainfall)                     1.442109e-01
## s(lag2_monthly_rainfall)                     1.010202e-01
## s(lag3_monthly_rainfall)                     5.493949e-02
## s(lag1_Relative_humidity)                    7.875839e-02
## s(lag2_Relative_humidity)                    7.447330e-02
## s(lag3_Relative_humidity)                    1.088194e-01
## s(yearmon)                                   1.328714e-01
##                                s(lag3_daytime_temperature)
## para                                         6.315086e-33
## s(daytime_temperature)                       1.093374e-01
## s(nighttime_temperature)                     1.086603e-01
## s(monthly_rainfall)                          1.136032e-01
## s(Relative_humidity)                         1.003212e-01
## s(lag1_daytime_temperature)                  1.044320e-01
## s(lag2_daytime_temperature)                  1.835780e-01
## s(lag3_daytime_temperature)                  1.000000e+00
## s(lag1_nighttime_temperature)                1.410313e-01
## s(lag2_nighttime_temperature)                2.125227e-01
## s(lag3_nighttime_temperature)                1.476569e-01
## s(lag1_monthly_rainfall)                     1.292164e-01
## s(lag2_monthly_rainfall)                     1.358905e-01
## s(lag3_monthly_rainfall)                     1.009755e-01
## s(lag1_Relative_humidity)                    1.034641e-01
## s(lag2_Relative_humidity)                    7.609345e-02
## s(lag3_Relative_humidity)                    7.031511e-02
## s(yearmon)                                   1.225962e-01
##                                s(lag1_nighttime_temperature)
## para                                           4.532024e-33
## s(daytime_temperature)                         1.090793e-01
## s(nighttime_temperature)                       2.571162e-01
## s(monthly_rainfall)                            1.463811e-01
## s(Relative_humidity)                           1.183478e-01
## s(lag1_daytime_temperature)                    1.597855e-01
## s(lag2_daytime_temperature)                    2.117244e-01
## s(lag3_daytime_temperature)                    1.281249e-01
## s(lag1_nighttime_temperature)                  1.000000e+00
## s(lag2_nighttime_temperature)                  2.482594e-01
## s(lag3_nighttime_temperature)                  1.479288e-01
## s(lag1_monthly_rainfall)                       1.704229e-01
## s(lag2_monthly_rainfall)                       1.217378e-01
## s(lag3_monthly_rainfall)                       8.429244e-02
## s(lag1_Relative_humidity)                      1.218207e-01
## s(lag2_Relative_humidity)                      1.232533e-01
## s(lag3_Relative_humidity)                      1.108690e-01
## s(yearmon)                                     7.147139e-02
##                                s(lag2_nighttime_temperature)
## para                                           1.100853e-32
## s(daytime_temperature)                         9.737075e-02
## s(nighttime_temperature)                       1.402915e-01
## s(monthly_rainfall)                            1.052880e-01
## s(Relative_humidity)                           1.024875e-01
```

```
## s(lag1_daytime_temperature)                    1.092227e-01
## s(lag2_daytime_temperature)                    1.615539e-01
## s(lag3_daytime_temperature)                    2.135857e-01
## s(lag1_nighttime_temperature)                  2.595959e-01
## s(lag2_nighttime_temperature)                  1.000000e+00
## s(lag3_nighttime_temperature)                  2.414054e-01
## s(lag1_monthly_rainfall)                       1.502259e-01
## s(lag2_monthly_rainfall)                       1.689198e-01
## s(lag3_monthly_rainfall)                       1.229875e-01
## s(lag1_Relative_humidity)                      1.200232e-01
## s(lag2_Relative_humidity)                      1.185263e-01
## s(lag3_Relative_humidity)                      1.206665e-01
## s(yearmon)                                     6.333986e-02
##                               s(lag3_nighttime_temperature)
## para                                           2.618112e-33
## s(daytime_temperature)                         1.020086e-01
## s(nighttime_temperature)                       1.113804e-01
## s(monthly_rainfall)                            1.041866e-01
## s(Relative_humidity)                           1.310823e-01
## s(lag1_daytime_temperature)                    9.667029e-02
## s(lag2_daytime_temperature)                    1.063069e-01
## s(lag3_daytime_temperature)                    1.573898e-01
## s(lag1_nighttime_temperature)                  1.370795e-01
## s(lag2_nighttime_temperature)                  2.532781e-01
## s(lag3_nighttime_temperature)                  1.000000e+00
## s(lag1_monthly_rainfall)                       1.063309e-01
## s(lag2_monthly_rainfall)                       1.497273e-01
## s(lag3_monthly_rainfall)                       1.676158e-01
## s(lag1_Relative_humidity)                      1.037638e-01
## s(lag2_Relative_humidity)                      1.173940e-01
## s(lag3_Relative_humidity)                      1.182865e-01
## s(yearmon)                                     5.928473e-02
##                               s(lag1_monthly_rainfall) s(lag2_monthly_rainfall)
## para                                     4.996842e-33             6.102804e-33
## s(daytime_temperature)                   5.946932e-02             8.051750e-02
## s(nighttime_temperature)                 1.133399e-01             7.849448e-02
## s(monthly_rainfall)                      7.873911e-02             8.196601e-02
## s(Relative_humidity)                     9.662954e-02             6.823094e-02
## s(lag1_daytime_temperature)              8.870389e-02             6.151899e-02
## s(lag2_daytime_temperature)              1.237992e-01             9.161122e-02
## s(lag3_daytime_temperature)              1.315328e-01             1.175481e-01
## s(lag1_nighttime_temperature)            1.483453e-01             1.118187e-01
## s(lag2_nighttime_temperature)            1.292025e-01             1.445921e-01
## s(lag3_nighttime_temperature)            1.046376e-01             1.313044e-01
## s(lag1_monthly_rainfall)                 1.000000e+00             7.996606e-02
## s(lag2_monthly_rainfall)                 9.213444e-02             1.000000e+00
## s(lag3_monthly_rainfall)                 8.611352e-02             8.704261e-02
## s(lag1_Relative_humidity)                1.454198e-01             9.716703e-02
## s(lag2_Relative_humidity)                1.036401e-01             1.445890e-01
## s(lag3_Relative_humidity)                9.082595e-02             1.025129e-01
## s(yearmon)                               9.638119e-02             8.931328e-02
##                               s(lag3_monthly_rainfall)
## para                                     4.017801e-33
## s(daytime_temperature)                   8.602039e-02
```

```
## s(nighttime_temperature)                 7.555818e-02
## s(monthly_rainfall)                      6.366710e-02
## s(Relative_humidity)                     7.546114e-02
## s(lag1_daytime_temperature)              7.903375e-02
## s(lag2_daytime_temperature)              6.274212e-02
## s(lag3_daytime_temperature)              8.628600e-02
## s(lag1_nighttime_temperature)            8.023019e-02
## s(lag2_nighttime_temperature)            1.106415e-01
## s(lag3_nighttime_temperature)            1.398039e-01
## s(lag1_monthly_rainfall)                 7.913454e-02
## s(lag2_monthly_rainfall)                 7.750030e-02
## s(lag3_monthly_rainfall)                 1.000000e+00
## s(lag1_Relative_humidity)                6.750430e-02
## s(lag2_Relative_humidity)                9.636392e-02
## s(lag3_Relative_humidity)                1.405050e-01
## s(yearmon)                               8.595880e-02
##                              s(lag1_Relative_humidity)
## para                                      4.800896e-33
## s(daytime_temperature)                    1.034138e-01
## s(nighttime_temperature)                  1.053791e-01
## s(monthly_rainfall)                       1.111078e-01
## s(Relative_humidity)                      1.336774e-01
## s(lag1_daytime_temperature)               7.726366e-02
## s(lag2_daytime_temperature)               8.255422e-02
## s(lag3_daytime_temperature)               1.163561e-01
## s(lag1_nighttime_temperature)             1.391055e-01
## s(lag2_nighttime_temperature)             1.356795e-01
## s(lag3_nighttime_temperature)             9.930996e-02
## s(lag1_monthly_rainfall)                  1.347032e-01
## s(lag2_monthly_rainfall)                  9.552126e-02
## s(lag3_monthly_rainfall)                  8.219780e-02
## s(lag1_Relative_humidity)                 1.000000e+00
## s(lag2_Relative_humidity)                 1.156288e-01
## s(lag3_Relative_humidity)                 9.426811e-02
## s(yearmon)                                1.159005e-01
##                              s(lag2_Relative_humidity)
## para                                      7.848537e-33
## s(daytime_temperature)                    1.111204e-01
## s(nighttime_temperature)                  8.281825e-02
## s(monthly_rainfall)                       8.975854e-02
## s(Relative_humidity)                      8.884104e-02
## s(lag1_daytime_temperature)               1.023717e-01
## s(lag2_daytime_temperature)               7.656760e-02
## s(lag3_daytime_temperature)               7.972317e-02
## s(lag1_nighttime_temperature)             1.053580e-01
## s(lag2_nighttime_temperature)             1.353668e-01
## s(lag3_nighttime_temperature)             1.302764e-01
## s(lag1_monthly_rainfall)                  1.080155e-01
## s(lag2_monthly_rainfall)                  1.321564e-01
## s(lag3_monthly_rainfall)                  9.740440e-02
## s(lag1_Relative_humidity)                 1.283755e-01
## s(lag2_Relative_humidity)                 1.000000e+00
## s(lag3_Relative_humidity)                 1.144767e-01
## s(yearmon)                                1.103483e-01
```

```
##                                s(lag3_Relative_humidity)   s(yearmon)
## para                                        2.700990e-32 2.320506e-32
## s(daytime_temperature)                      9.412489e-02 9.702456e-02
## s(nighttime_temperature)                    1.159030e-01 5.905813e-02
## s(monthly_rainfall)                         7.537228e-02 7.432394e-02
## s(Relative_humidity)                        9.003193e-02 9.050082e-02
## s(lag1_daytime_temperature)                 1.086516e-01 9.894708e-02
## s(lag2_daytime_temperature)                 1.028073e-01 1.015844e-01
## s(lag3_daytime_temperature)                 7.584376e-02 1.017293e-01
## s(lag1_nighttime_temperature)               8.049952e-02 6.196675e-02
## s(lag2_nighttime_temperature)               1.026464e-01 6.089326e-02
## s(lag3_nighttime_temperature)               1.304566e-01 5.978974e-02
## s(lag1_monthly_rainfall)                    9.803348e-02 7.654505e-02
## s(lag2_monthly_rainfall)                    1.060342e-01 7.402087e-02
## s(lag3_monthly_rainfall)                    1.287822e-01 7.209209e-02
## s(lag1_Relative_humidity)                   8.773224e-02 8.669201e-02
## s(lag2_Relative_humidity)                   1.254598e-01 8.236149e-02
## s(lag3_Relative_humidity)                   1.000000e+00 7.825673e-02
## s(yearmon)                                  1.055380e-01 1.000000e+00
```

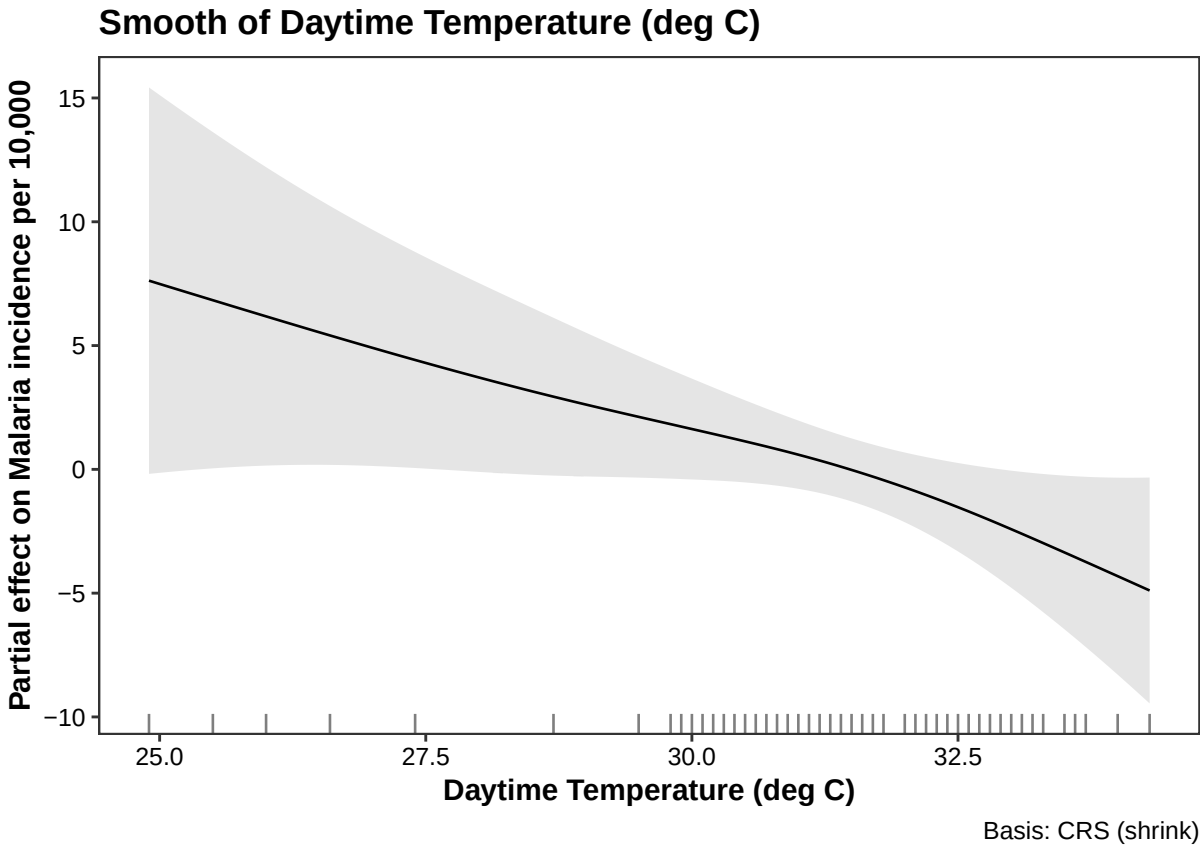
7 Smooth-Term Plots

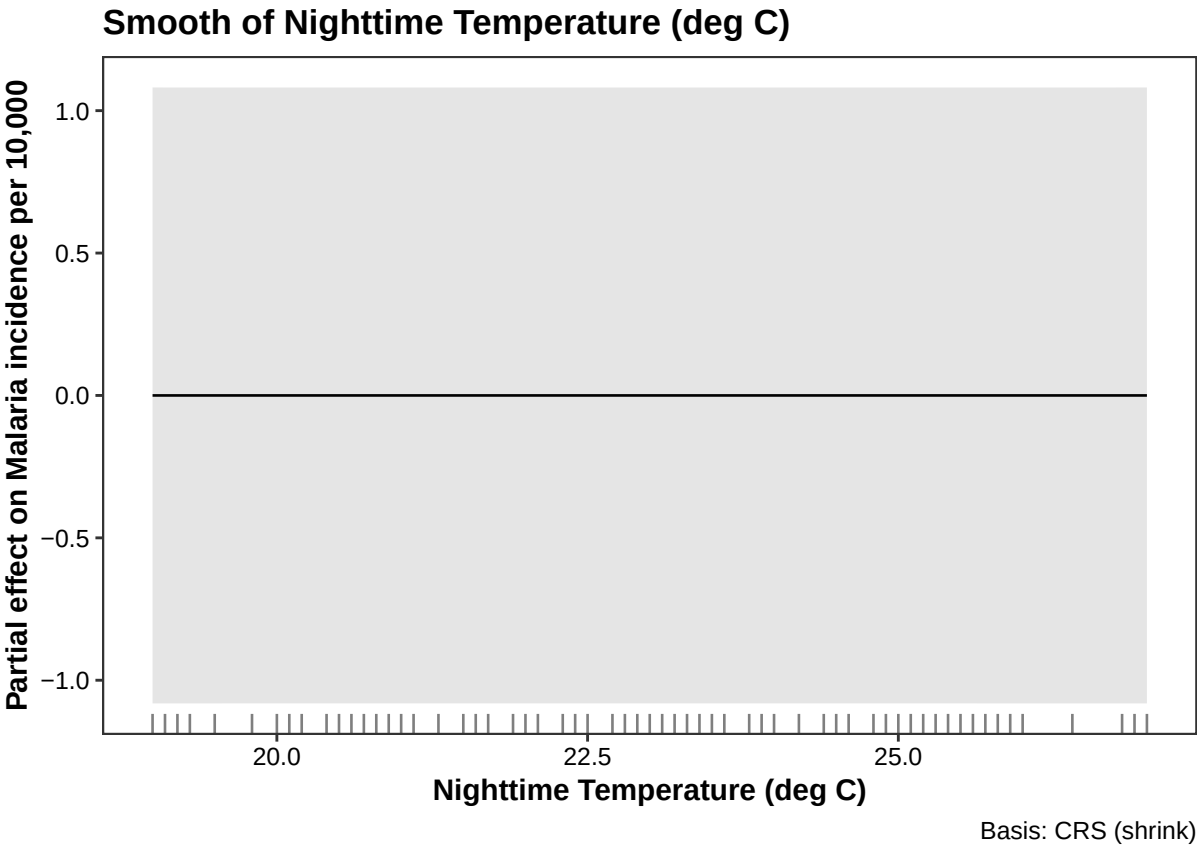
```r
sm_names <- gratia::smooths(gam_fit)
stopifnot(is.character(sm_names) && length(sm_names) > 0)

purrr::walk2(
  sm_names,
  seq_along(sm_names),
  ~{
    p <- draw_one_smooth(gam_fit, .x)
    base <- sprintf("%02d_%s", .y, clean_name(.x))
    ggplot2::ggsave(
      filename = file.path(plots_dir, paste0(base, ".png")),
      plot = p, width = 6.5, height = 5, units = "in", dpi = 300
    )
  }
)

cat("Saved PNG plots to:", plots_dir, "\n")
```

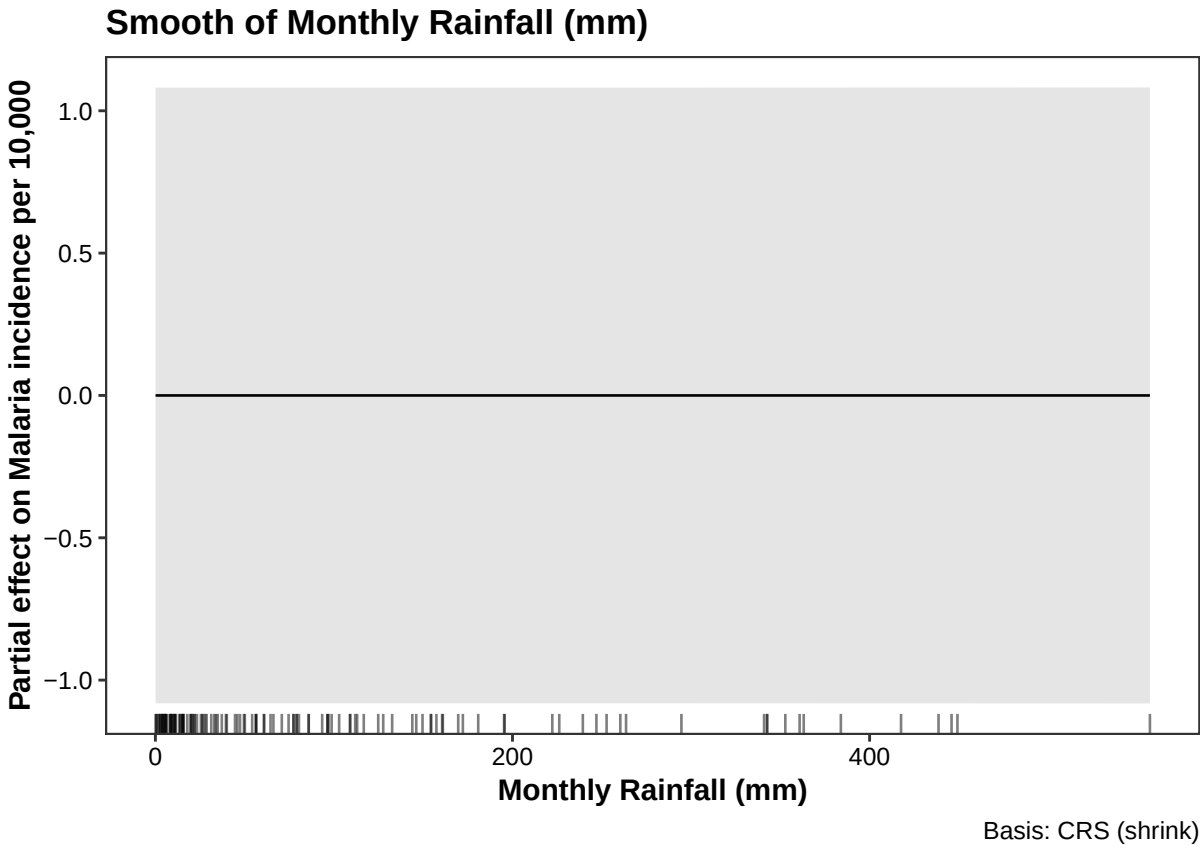
```
## Saved PNG plots to: C:/Users/USER/Videos/Climate and Malaria Plots and Analysis/Smooth Term Plots
```

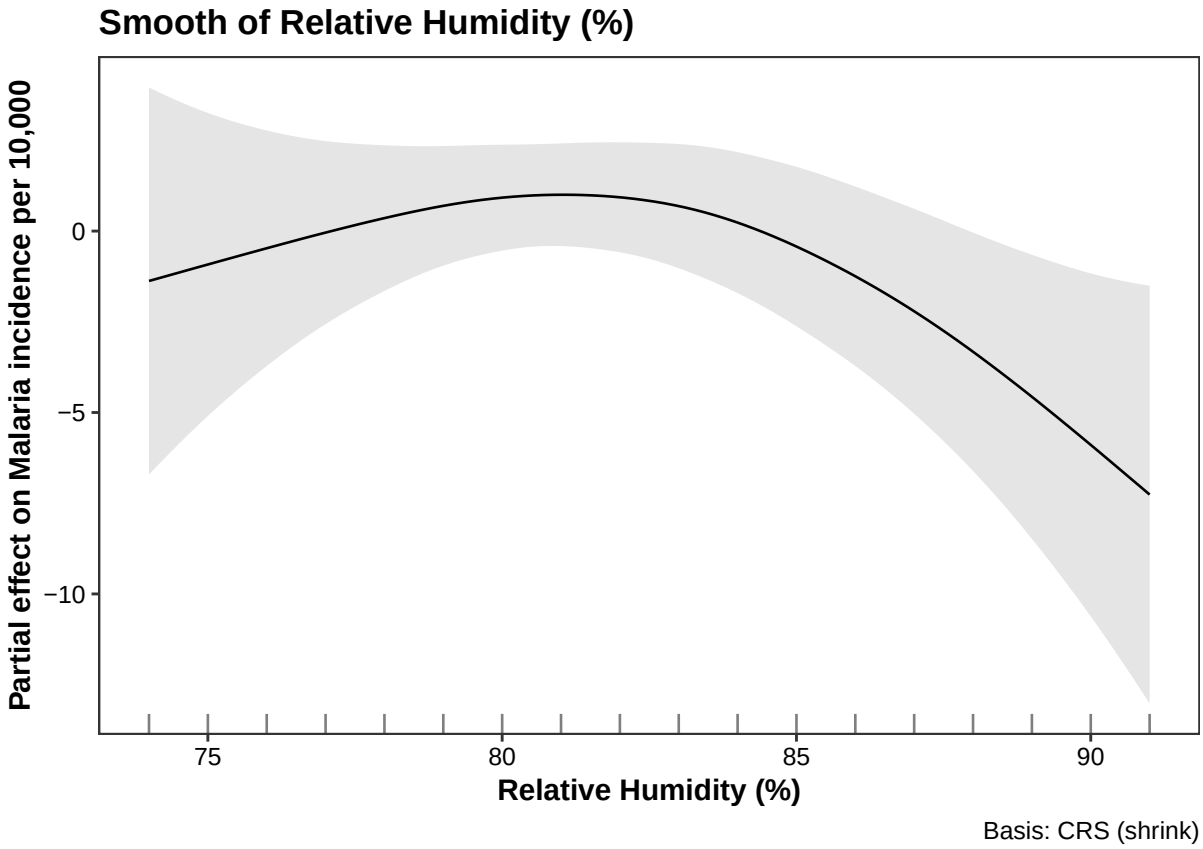
7.1 Example Smooths (inline)

```r
preview_n <- min(4, length(sm_names))
for (nm in sm_names[seq_len(preview_n)]) {
  print(draw_one_smooth(gam_fit, nm))
}
```


Smooth of Daytime Temperature (deg C)
Partial effect on Malaria incidence per 10,000
15
10
5
0
−5
−10
25.0
27.5
30.0
32.5
Daytime Temperature (deg C)
Basis: CRS (shrink)


Smooth of Nighttime Temperature (deg C)
Partial effect on Malaria incidence per 10,000
1.0
0.5
0.0
−0.5
−1.0
20.0
22.5
25.0
Nighttime Temperature (deg C)
Basis: CRS (shrink)


Smooth of Monthly Rainfall (mm)
Partial effect on Malaria incidence per 10,000
1.0
0.5
0.0
−0.5
−1.0
0
200
400
Monthly Rainfall (mm)
Basis: CRS (shrink)


Smooth of Relative Humidity (%)
Partial effect on Malaria incidence per 10,000
0
−5
−10
75
80
85
90
Relative Humidity (%)
Basis: CRS (shrink)